

[TODO: TITLE] The $(s - 1)$ -adic valuation of Hikita's $\star$ -product: a block law, cumulant leads, and a box symmetry

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Abstract. **[TODO: Abstract (≤ 150 words, no custom macros, no cite).]** Hikita introduced a two-parameter deformation $\star$ of the product of symmetric functions. Writing $e_{\lambda_1} \star \cdots \star e_{\lambda_\ell} = \sum_{\mu} c_{\lambda\mu}(s, t) e_{\mu}$, we show that the $(s - 1)$ -adic valuation of $c_{\lambda\mu}$ is $\ell(\lambda) - \kappa(\lambda, \mu)$, where κ is the maximal number of blocks into which λ and μ split compatibly. The leading coefficients factor over block decompositions into connected pieces; for a full merge the connected piece is a connected-graph cumulant with edge weights $t^{\lambda_i \lambda_j} - 1$. The coefficients are invariant, up to a power of s , under complementation in a box. All second-order leading coefficients are given in closed form.

Keywords: Hikita star product, Hall–Littlewood functions, elementary symmetric functions, cumulants, valuations

1 Introduction

Hikita [8] defined a commutative, associative product $\star$ on $\Lambda_{q,t}$ by transporting the ordinary product through the level-one polynomial representation of the affine Hecke algebra; we always say *Hikita's* $\star$. We write $s := q^{-1}$. At $s = 1$, $\star$ is the ordinary product, and at $s = 0$ (after rescaling) it degenerates to a Hall–Littlewood product (Remark 2.3). Di Francesco and Kedem [3] observed that the associated t -deformed graded characters lose Schur positivity at generic t , and positivity of $\star$ -structure constants fails in every natural basis we tried. *Positivity is lost, but structure is not.* This abstract shows that the structure survives in the $(s - 1)$ -adic filtration.

We expand the structure constants of $e_{\lambda}^{\star} := e_{\lambda_1} \star \cdots \star e_{\lambda_\ell} = \sum_{\mu} c_{\lambda\mu}(s, t) e_{\mu}$ around $s = 1$. The point of departure is Hikita's formula for $e_k \star (\cdot)$ as a sum over k -subsets (Proposition 2.1). In it, s enters only through the Euler operator s^{Δ_A} , so the $(s - 1)^p$ Taylor piece of $e_k \star (\cdot)$ is a differential operator of order p . An order- p operator can glue at most p existing blocks of parts to a new part. Counting the gluings that the e -expansion forces gives a lower bound on the valuation. Matching it exactly is a cancellation question, and

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we settle it at $t = 0$, where $e_\lambda^\star$ is a modified Hall–Littlewood function and Macdonald’s raising operators make every gluing cost exactly one factor $(s - 1)$ with a fixed sign. Main results:

- **Block valuation law** (Theorem 3.4): $v_{s-1}(c_{\lambda\mu}) = \ell(\lambda) - \kappa(\lambda, \mu)$ for all $\mu \succeq \lambda$, $\mu \neq \lambda$, where κ is the maximal number of compatible blocks. The first-order term is a symmetric biderivation with an explicit matrix (Theorem 3.2 and Proposition 3.3).
- **Merge weights and full-merge leads** (Theorems 3.8 and 4.1): the leading coefficient for $\mu = (n)$ is $\frac{1-t^n}{\prod_i (1-t^{\lambda_i})} \sum_{H \text{ connected}} \prod_{ij \in H} (t^{\lambda_i \lambda_j} - 1)$. The graph identity is due to Dołęga [4]; the identification with $\star$ is ours.
- **Block multiplicativity** (Theorems 4.4 and 4.5): every leading coefficient is a sum over block decompositions of products of connected leading coefficients (exp of connected).
- **Box Complement** (Theorem 5.1): $c_{N^\ell - \lambda, N^\ell - \mu} = s^{N \binom{\ell}{2} - (\ell-1)|\lambda|} c_{\lambda\mu}$, together with a Column Lemma.
- **Second order** (Theorem 6.6 and Corollary 6.7): every leading coefficient at valuation 2 is in closed form, via a constant-term formula and two-part Green polynomials [6, 1, 12].

The leads have a clean hierarchy. At valuation 1 they are governed by pairing with p_n , and Lemma 3.9 evaluates that pairing by a principal specialisation, i.e. one t -string $1, t, \dots, t^{k-1}$. At valuation 2 they are governed by pairing with $p_x p_y$, and the poles of a constant-term integral fall into two interacting t -strings (Theorem 6.3). The full-merge lead is a cumulant: up to a prefactor it is the connected part of the character $e_k \mapsto t^{\binom{k}{2}}$. This places $\star$ next to the Macdonald cumulants of Dołęga [4] and Haglund–Tewari [7]. Their objects and ours share the graph half (Theorem 4.1 and the remark after it), but a gauge-invariant ratio separates $\star$ from Dołęga’s $\oplus$ -cumulants away from $t = 0$.

Organisation. Section 2 recalls Hikita’s product and the dominance support. Section 3 proves the block valuation law and the merge weights. Section 4 treats leading coefficients, Section 5 the Box Complement theorem and the linear coefficient, and Section 6 the second order. Proofs are sketched here; full proofs will appear in a longer version.

2 Background: Hikita’s product and Hall–Littlewood notation

Notation. $\Lambda = \mathbb{Q}(t)[e_1, e_2, \dots]$; $I = (e_1, e_2, \dots)$ is the augmentation ideal, so $I^b = \text{span}\{e_\mu : \ell(\mu) \geq b\}$. For G homogeneous of degree n , $\text{lin}_e G := [e_n]G$. $[m] = [m]_t =$

$(1 - t^m)/(1 - t)$, $\varphi_m = \prod_{i=1}^m (1 - t^i)$, $n(\lambda) = \sum_i (i - 1)\lambda_i$, $\supseteq$ is dominance. $P_\lambda(x; t)$, $Q_\lambda = b_\lambda P_\lambda$, Q'_λ are the Hall–Littlewood functions of [12, Ch. III], with the Hall pairing $\langle \cdot, \cdot \rangle$ and the HL pairing $\langle p_\lambda, p_\mu \rangle_t = \delta_{\lambda\mu} z_\lambda \prod_i (1 - t^{\lambda_i})^{-1}$.

Hikita's product. $F \star G := q(q^{-1}F \cdot q^{-1}G)$, where q is the level-one polynomial representation map $F(Y) \mapsto F(Y) \bullet 1$ of the affine Hecke algebra of GL_m , stable in m [8, §3.1, Def. 3.4, Cor. 3.10]. We use $s = q^{-1}$; then Hikita's Pieri rule [8, Thm. 3.12] reads $e_1 \star e_r = (1 - s)[r + 1]e_{r+1} + s e_1 e_r$. Here $[m] = [m]_t$.

Proposition 2.1 (Subset formula). *For $F \in \Lambda_N = \mathbb{Q}(t)[x_1, \dots, x_N]^{S_N}$, $N \geq \deg$,*

$$e_k \star F = E_k F := \sum_{|A|=k} c_A X_A F(X_{A^c}, sX_A), \quad c_A = \prod_{i \in A, j \notin A} \frac{x_i - tx_j}{x_i - x_j}, \quad X_A = \prod_{i \in A} x_i.$$

Proof idea. By [8, Def. 3.4, Lemma 3.3], $e_k \star F = t^{-\binom{k}{2}} e_k(Y) \bullet F$, with Y_i the Cherednik operators in N variables. Write $e_k(Y)$ with increasing indices, and let $\pi F = x_1 F(x_2, \dots, x_N, sx_1)$. A Coxeter length count on minimal coset representatives of $S_k \times S_{N-k}$ gives $e_k(Y)F = t^{\binom{k}{2}} \sum_{|D|=k} T_{w^D} \pi^k F$. For G symmetric in the first k and the last $N - k$ variables, the chain factorisation of this coset sum and the Poincaré identity $\sum_{w \in S_k} w(\prod_{i < j} \frac{x_i - tx_j}{x_i - x_j}) = [k]_t!$ give $\sum_D T_{w^D} G = \sum_{|A|=k} c_A G^{(A)}$, where $G^{(A)}$ places the head variables on A . $\square$

Write $e_\lambda^\star := E_{\lambda_1} \cdots E_{\lambda_\ell}(1) = \sum_\mu c_{\lambda\mu}(s, t) e_\mu$. The variable s enters only through $F(X_{A^c}, sX_A) = s^{\Delta_A} F$, $\Delta_A = \sum_{i \in A} x_i \partial_i$; Taylor expansion at $s = 1$ gives

$$E_k = \sum_{p \geq 0} (s - 1)^p E_k^{(p)}, \quad E_k^{(p)} = \sum_{|A|=k} c_A X_A \binom{\Delta_A}{p}, \quad (2.1)$$

and $E_k^{(p)}$ is a differential operator of order $\leq p$ on Λ_N , with $E_k^{(0)} = e_k \cdot$ and $E_k^{(p)}(1) = 0$ for $p \geq 1$.

Theorem 2.2 (Dominance support; the $s = 0$ valuation). $e_\lambda^\star = s^{n(\lambda)} e_\lambda + \sum_{\mu \triangleright \lambda} c_{\lambda\mu} e_\mu$ with $c_{\lambda\mu} \in \mathbb{Q}[s, t]$ and $c_{\lambda\mu}|_{s=1} = 0$. Moreover $c_{\lambda\mu} \neq 0$ for every $\mu \supseteq \lambda$, and the s -adic valuation of $c_{\lambda\mu}$ is exactly $n(\mu)$. At $s = t = 0$, in the basis $b_\mu = s^{n(\mu)} e_\mu$, $e_\lambda^\star$ is the zeta function of dominance order.

Proof idea. A termwise degree count on the subset formula: for symmetric P let $D_j(P)$ be the maximal total degree in j variables; E_k raises D_j by at most $\min(k, j)$, and $D_j(P) \leq \sum_i \min(\rho_i, j)$ for all j characterises $\text{span}\{e_\nu : \nu \supseteq \rho\}$ [12, I (1.11), (6.6)]. **[TODO: Out of v7.5 scope: keep as background only, or cut to one sentence.]** $\square$

Remark 2.3 (The $s \rightarrow 0$ edge; no novelty claim for the matrix). Since $s = q^{-1}$, the edge $s \rightarrow 0$ is Hikita's $q \rightarrow \infty$, where the unrescaled product degenerates [8, Lemma 6.3]. In the basis b_μ , $e_k \star (\cdot)$ preserves $\bigoplus_\mu \mathbb{Q}[s, t] b_\mu$, and modulo s it is conjugate, via $b_\mu \mapsto t^{-n(\mu')} P_{\mu'}(x; t^{-1})$, to multiplication by $t^{-\binom{k}{2}} e_k$. Consequently the leading coefficients $d_{\lambda\mu}(t) := [s^{n(\mu)}] c_{\lambda\mu}$ of [Theorem 2.2](#) are the known $e \rightarrow \text{HL}$ transition matrix of [11, §3.2]: writing $e_\lambda = \sum_\kappa M(e, P)_{\lambda\kappa} P_\kappa(x; t)$,

$$d_{\lambda\mu}(t) = t^{n(\mu') - n(\lambda')} M(e, P)_{\lambda\mu'}(t^{-1}).$$

Their positivity and $d_{\lambda\mu}(1) = \#\{0\text{-}1 \text{ matrices with row sums } \lambda, \text{ column sums } \mu'\}$ are therefore classical [11, §3.2]; what is new is only the identification of this matrix with the s -leading coefficients of $\star$.

3 The block valuation law at $s = 1$

Definition 3.1. For $\lambda, \mu \vdash n$, $\kappa(\lambda, \mu)$ is the largest r such that $\lambda = \lambda^1 \sqcup \dots \sqcup \lambda^r$ and $\mu = \mu^1 \sqcup \dots \sqcup \mu^r$ as multisets of parts, with every block nonempty and $\mu^i \supseteq \lambda^i$ (so $|\mu^i| = |\lambda^i|$); $\kappa = -\infty$ if $\mu \not\supseteq \lambda$. μ is a *coarsening* of λ if $\kappa(\lambda, \mu) = \ell(\mu)$. v denotes the $(s - 1)$ -adic valuation.

Theorem 3.2 (First order is a biderivation). *Let $D_k := E_k^{(1)}$ and $B(f, g) := \partial_s(f \star g)|_{s=1}$. Then D_k is a derivation of Λ_N , and*

$$B(f, g) = \sum_{k,l} M_{kl} \frac{\partial f}{\partial e_k} \frac{\partial g}{\partial e_l}, \quad M_{kl} := D_k(e_l) = B(e_k, e_l) = M_{lk}.$$

So B is a symmetric biderivation, the carré du champ of $L = \frac{1}{2} \sum M_{kl} \partial^2 / \partial e_k \partial e_l$.

Proof idea. $(\Delta_A^1) = \Delta_A$ is a derivation and left multiplication by $c_A X_A$ preserves this. Differentiate $f \star g = \sum_\rho a_\rho(s) E_\rho g$ at $s = 1$ and use $D_k(1) = 0$. Symmetry of M is commutativity of $\star$ [8]. $\square$

Proposition 3.3 (The matrix M). *For $k \geq r \geq 0$, with $L(a, b) := \frac{(1 - t^{a+b})(t^{ab} - 1)}{(1 - t^a)(1 - t^b)}$,*

$$M_{kr} = B(e_k, e_r) = r e_k e_r + \sum_{j=1}^r L(k - r + j, j) e_{k+j} e_{r-j}.$$

Consequently $[(s - 1)^1] e_\lambda^\star = \sum_{i < j} M_{\lambda_i \lambda_j} e_{\lambda \setminus \{\lambda_i, \lambda_j\}}$.

Proof idea. Differentiate [Corollary 5.5](#) (whose proof does not use [Proposition 3.3](#)) at $s = 1$. Since $\pi_a(b)|_{s=1} = \delta_{b0}$, only $y = r$ survives at $s = 1$, and its derivative is $r e_k e_r$. For $y = r - j$, $j \geq 1$, put $a = k - r + j$. Then $\partial_s \pi_a(j)|_{s=1} = [u^j] \sum_{m < a} \frac{u t^m}{1 + u t^m} = (-1)^{j-1} [a]_{t^j}$, so $\partial_s C_{a,j}|_{s=1} = -\frac{[a+j]}{[a]} [a]_{t^j} = L(a, j)$. The second statement follows from [Theorem 3.2](#), since $D_k(1) = 0$. $\square$

Theorem 3.4 (Block valuation law). *For all $\lambda, \mu \vdash n$ and all t , $v(c_{\lambda\mu}) \geq \ell(\lambda) - \kappa(\lambda, \mu)$. For $\mu \succeq \lambda$, $\mu \neq \lambda$, equality holds over $\mathbb{Q}(t)$ and at $t = 0$:*

$$v(c_{\lambda\mu}) = \ell(\lambda) - \kappa(\lambda, \mu).$$

At $t = 0$ the leading coefficient is $(-1)^{\ell-\kappa} N(\lambda, \mu)$ with $N(\lambda, \mu) \geq 1$ a count of minimal raising-operator configurations.

Proof idea. Lower bound (subset formula only): by polarization, an order- p operator glues at most p existing “blocks” to the new part, so the $(s-1)^m$ coefficient is a sum of products over $\geq \ell - m$ blocks, each of type λ_C by the degree count of [Theorem 2.2](#). Equality: at $t = 0$, $e_\lambda^* = \omega \tilde{H}_\lambda(x; s)$; Macdonald’s raising operators $Q'_\lambda = \prod_{i < j} \frac{1 - R_{ij}}{1 - q R_{ij}} h_\lambda$ [[12](#), III (2.15)] make every transfer edge cost exactly one $(s-1)$ with sign -1 , so nothing cancels; the minimum edge count is $\ell - \kappa$ (components $\leq \kappa$; a path tree on each indecomposable block). Specialise $t = 0 \rightarrow$ generic. $\square$

Remark 3.5 (Input at $t = 0$; (N)-freeness). The $t = 0$ edge $e_\mu^*|_{t=0} = s^{n(\mu)} P_{\mu'}(x; 1/s, 0) = \omega \tilde{H}_\mu(x; s)$ is **not ours**: it is assembled from [[2](#)] eq. (5.15) (with $M_{k,1} = E_k|_{t=0}$), Cor. 5.8 in its level-one form (5.25), Cor. 5.18 (eq. (5.27)) (arXiv v2 numbering), and [[12](#), VI (5.1), (4.14)(iv)]. In particular [Theorem 3.4](#) does not depend on the ∇ -transport or on Cherednik’s nonsymmetric theory; the theorem numbers above are those of arXiv:1505.01657v2.

Remark 3.6 (Novelty). The lower bound and the generic- t statement are novel-as-checked but formally **UNAUDITED** beyond the $t = 0$ case. The $t = 0$ case is likely folklore ($\sim 75\%$): it follows in ~ 10 lines from [[1](#)] eq. (11) plus Jacobi–Trudi; no paper states it. Present the $t = 0$ case as a classical-style exercise. **[TODO: Kirillov–Noumi / Chen–Lu–Ruan 2601.13497 Cor. 2.10 gates listed in v7.2.]**

Theorem 3.7 (Coarsenings: merge-history expansion). *Let $\mu \neq \lambda$ be a coarsening of λ and $m = \ell(\lambda) - \ell(\mu)$. Then $v(c_{\lambda\mu}) = m$, and $\text{Lead}_{\lambda\mu}(t) := [(s-1)^m] c_{\lambda\mu}$ equals*

$$\text{Lead}_{\lambda\mu}(t) = \sum_{\text{tight histories } H \text{ ending at } \mu} \prod_{\text{merges in } H} W_k(J), \quad \text{Lead}_{\lambda\mu}(0) = (-1)^{m\#\{H\}} \neq 0,$$

where a history processes $\lambda_\ell, \dots, \lambda_1$, each part k either opening a block or merging $p \geq 1$ existing blocks of sizes $J = (j_1, \dots, j_p)$, and $W_k(J) := \text{lin}_e [\dots [E_k^{(p)}, e_{j_1}], \dots, e_{j_p}](1)$.

Proof idea. Pass to $\text{gr}_I \Lambda$: non-tight terms land in $I^{\ell-m+1}$; tight steps are multilinear derivations, so only linear parts $g_i \equiv \gamma_i e_{n_i} \bmod I^2$ matter. At $t = 0$, a roots-of-unity evaluation, Weyl orthogonality and Murnaghan–Nakayama give $W_k(J)|_{t=0} = (-1)^p$. This proof does not use (N). $\square$

Theorem 3.8 (Merge weights). *With $n = k + |J|$ and $p = |J|$,*

$$W_k(J) = (-1)^p \frac{[n]_t}{[k]_t} \prod_{j \in J} [k]_{t^j}.$$

Consequently $(-1)^m \text{Lead}_{\lambda_\mu}(t) > 0$ for coarsenings and every real $t > 0$.

Proof idea. $W_k(J) = (-1)^{d-p} \text{lin}_e T_k(p_J)$ with $T_k g := \sum_{|A|=k} c_A X_A g(X_A)$. The parabolic factorisation $T_k P_\rho = P_{\rho+1^k}$ [12, III (2.2)] and $\text{lin}_e P_\lambda = (-1)^{n-k} \frac{[n]_t}{[k]_t} P_{\lambda-1^k}(1, t, \dots, t^{k-1})$ (from [12, III.7 Ex. 2, III.2 Ex. 1]) give the “discrete HL measure = principal specialisation” identity $\text{lin}_e T_k f = (-1)^d \frac{[n]_t}{[k]_t} f(1, t, \dots, t^{k-1})$ (Lemma 3.9); apply it to $f = p_J$. $\square$

Lemma 3.9. *For $f \in \Lambda_k$ homogeneous of degree d and $n = k + d$: $\text{lin}_e T_k(f) = (-1)^d \frac{[n]_t}{[k]_t} f(1, t, \dots, t^{k-1})$.*

Remark 3.10 (Novelty). Lemma 3.9 is a short corollary of [12, III.7 Ex. 2, III (2.2)]; likely folklore-level, do not headline. Theorem 3.8’s cold recheck was done (Day 223 §4). The proof sketched above avoids the Hall–Littlewood expansion of E_k used in the first proof, but both proofs evaluate $\text{lin}_e P_\lambda$ by the same lemma, so they are not independent.

4 Leading coefficients: connected-graph cumulants and block multiplicativity

Theorem 4.1 (Full-merge lead). *For $\lambda \vdash n$ with $\ell = \ell(\lambda) \geq 2$,*

$$\text{Lead}_{\lambda, (n)}(t) = \frac{1 - t^n}{\prod_i (1 - t^{\lambda_i})} K_\lambda(t), \quad K_\lambda(t) := \sum_{H \text{ connected on } [\ell]} \prod_{ij \in H} (t^{\lambda_i \lambda_j} - 1).$$

K_λ is the connected part (joint cumulant) of $t^{e_2(\lambda)} = \prod_{i < j} (1 + (t^{\lambda_i \lambda_j} - 1))$.

Proof idea. (1) Tight histories ending in one block are increasing trees on $[\ell]$ (parent = the merging part). (2) With N_v the λ -mass of the subtree at v , Theorem 3.8 telescopes along the tree to $\frac{1-t^n}{\prod(1-t^{\lambda_i})} \prod_{v \rightarrow c} (t^{\lambda_v N_c} - 1)$. (3) Tree-graph identity: $\sum_{\text{incr. trees}} \prod_{v \rightarrow c} (\prod_{j \in T_c} x_{vj} - 1) = \sum_{\text{conn. } H} \prod_{e \in H} (x_e - 1)$ for generic x , by the minimum-vertex/component decomposition; specialise $x_{ij} = t^{\lambda_i \lambda_j}$. $\square$

Corollary 4.2. (a) $\lambda = 1^n$: $\text{Lead} = (-1)^{n-1} [n]_t I_n(t)$ with I_n the tree inversion enumerator [13]. (b) $t = 0$: $\text{Lead} = (-1)^{\ell-1} (\ell-1)!$. (c) $t \rightarrow 1$: $\text{Lead} \rightarrow (-1)^{\ell-1} n^{\ell-1}$ (weighted Cayley). (d) $\text{sgn Lead} = (-1)^{\ell-1}$ for all $t > 0, t \neq 1$.

Proof of (b), (c). At $t = 0$ every edge weight is -1 and $\sum_{H \text{ conn.}} (-1)^{|H|} = (-1)^{\ell-1} (\ell-1)!$. As $t \rightarrow 1$, with $\varepsilon = t - 1$, only spanning trees contribute to the $\varepsilon^{\ell-1}$ term of K_λ , and $\sum_T \prod_{i,j \in T} \lambda_i \lambda_j = (\prod_i \lambda_i) n^{\ell-2}$ (weighted Cayley); the prefactor is $(-n\varepsilon) / ((-\varepsilon)^\ell \prod \lambda_i) \cdot (1 + O(\varepsilon))$. $\square$

Remark 4.3 (Prior art — graph half). The tree, connected-graph and cumulant forms of K_λ are a special case of Dołęga [4, Prop. 2.1 and (11), Lemma 2.3] (Tutte polynomial $T(1, t)$ of the graph M_λ), cf. also [5, 13]; step (3) is a Penrose-type tree-graph identity [15]. **Must cite, not claim.** Ours: the identification of the $\star$ -lead with prefactor $\cdot K_\lambda$ (Theorems 3.7, 3.8 and 4.1) and Theorems 4.4 and 4.5. Haglund–Tewari’s single-row cumulant [7, Def. 7.1, Thm. 7.3] (their q is our t) is the same graph half: since $\langle \tilde{H}_{(m)}, e_m \rangle = q^{\binom{m}{2}}$, one has $\langle \kappa(\lambda), e_n \rangle = t^{n(\lambda')} (t-1)^{1-\ell} K_\lambda(t)$. *Separator.* Full-merge leads $L(\lambda)$ of any such theory are only defined up to the gauge $L(\lambda) \mapsto L(\lambda) a^{\ell(\lambda)} g(|\lambda|) \prod_i f(\lambda_i)$ (rescaling generators, degree-wise normalisation, rescaling the expansion parameter). The first gauge invariant appears at $n = 4$: $J := L(1^4) L(2, 2) / L(2, 1, 1)^2$. For $\star$, Theorem 4.1 gives $J = \frac{(t^2+1)(t^3+3t^2+6t+6)}{(t+1)(t^2+t+2)^2}$, and Haglund–Tewari’s $\langle \kappa, e_n \rangle$ gives the same value. Dołęga’s $\oplus$ -cumulant of columns (lowest $(q-1)$ -coefficient of $[g_n] \tilde{H}_{\lambda'}, g_k = \tilde{H}_{1^k}$) gives $J \equiv 3/2 = J_\star(0)$. So Dołęga’s $\oplus$ agrees with $\star$ only at $t = 0$ (Remark 3.5), and Haglund–Tewari’s is the same graph object without $\star$.

Theorem 4.4 (Coarsening leads factor). For μ a coarsening of λ ,

$$\text{Lead}_{\lambda\mu} = \sum_{\pi} \prod_{B \in \pi} \text{Lead}_{\lambda_B, (|\lambda_B|)},$$

summed over set partitions π of the positions $[\ell]$ whose sorted block sums equal μ (singletons contribute 1). There is no interleaving factor.

Proof idea. Increasing forests = set partition $\times$ increasing tree per block, and the telescoped weight is local to each tree. $\square$

Theorem 4.5 (Block multiplicativity). Let $\kappa = \kappa(\lambda, \mu) \geq 1$, $\ell = \ell(\lambda)$, $m = \ell - \kappa$. Then

$$[(s-1)^m] c_{\lambda\mu} = \sum_{\pi} \sum_{(v^C)_{C \in \pi}} \prod_{C \in \pi} [(s-1)^{|C|-1}] c_{\lambda_C, v^C},$$

where π runs over set partitions of $[\ell]$ into exactly κ blocks and (v^C) over tuples with $\bigsqcup_C v^C = \mu$ and $v^C \supseteq \lambda_C$; every factor then has $\kappa(\lambda_C, v^C) = 1$. With Theorem 3.4 this is $\text{Lead}_{\lambda\mu}$.

Proof idea. Expand by polarization into histories; non-tight histories produce $> \kappa$ compatible blocks and vanish by maximality of κ ; tight histories factor block by block and interleave uniquely. Uses commutativity of $\star$ [8] only to pass from compositions to partitions. $\square$

Remark 4.6. [Theorem 4.4](#) is the case where each ν^C is a single part. The shape is that of a Mayer/cluster expansion; the $\star$ -specific content is the connected-lead input. Novelty: unchecked as a separate statement [\[TODO: one search\]](#).

5 Box Complement and the linear coefficient

Theorem 5.1 (Box Complement). *Let λ, μ have at most ℓ parts (padded with zeros) and all parts $\leq N$; put $N^\ell - \lambda := (N - \lambda_\ell, \dots, N - \lambda_1)$. Then*

$$c_{N^\ell - \lambda, N^\ell - \mu}(s, t) = s^{N \binom{\ell}{2} - (\ell-1)|\lambda|} c_{\lambda\mu}(s, t).$$

Proof idea. Stability of the subset formula for every N (also $N < \text{degree}$), and an inversion lemma in N variables: for $F = e_N^m G(1/x)$ with G of degree g , $E_{N-k} F = s^{m(N-k)-g} e_N^{m+1}(E_k G)(1/x)$ (substitute $y = 1/x$, $B = A^c$). Iterate and compare coefficients using $e_{N-r}(x) = e_N(x) e_r(1/x)$ and [Theorem 2.2](#) (support has $\leq \ell$ parts). $\square$

Corollary 5.2 (Column Lemma; lead invariance). $c_{\lambda+1^\ell, \mu+1^\ell} = s^{\binom{\ell}{2}} c_{\lambda\mu}$. Hence v and Lead are invariant under $(\lambda, \mu) \mapsto (\lambda + 1^\ell, \mu + 1^\ell)$ and $(\lambda, \mu) \mapsto (N^\ell - \lambda, N^\ell - \mu)$; e.g. $(2, 2, 2) \rightarrow (3, 3)$ and $(2, 2, 2) \rightarrow (4, 1, 1)$ have the lead $[3]_t(t+2)$ of $(1, 1, 1) \rightarrow (3)$.

Remark 5.3 (Prior art). Mechanism: the inversion lemma follows in ~ 4 lines from [3, Rem. 3.3] ($S\Gamma_I S = \Gamma_I^{-1}$ under $x \mapsto x^{-1}$) plus subset complement; stability is [8, Prop. 3.8, Cor. 3.10]. Present as “Lemma (cf. DFK17 Rem. 3.3)”. The statement for $\star$ structure constants and [Corollary 5.2](#) were not found anywhere checked: novel-as-checked (wake 225 audit).

Theorem 5.4 (Plethystic linear coefficient). *For G homogeneous of degree d : $\text{lin}_e(e_k \star G) = (-1)^d \frac{[k+d]_t}{[k]_t} G[(s-1)[k]_t]$, where $G \mapsto G[(s-1)[k]_t]$ is the ring map $p_r \mapsto (s^r - 1)[k]_{t^r}$.*

Corollary 5.5 (Pieri rule, all s). $e_k \star e_r = \sum_{y=0}^{\min(k,r)} s^y C_{k-y, r-y} e_{k+r-y} e_y$ with $C_{a,0} = C_{0,b} = 1$, $C_{a,b} = (-1)^b \frac{[a+b]}{[a]} \pi_a(b)$, $\pi_a(j) = [u^j] \prod_{m=0}^{a-1} \frac{1+su t^m}{1+u t^m}$.

Proof idea. Support by [Theorem 2.2](#); strip y columns by [Corollary 5.2](#) (valid for compositions) to reduce to $\text{lin}_e E_{k-y}(e_{r-y})$; evaluate by [Theorem 5.4](#), which itself is [Lemma 3.9](#) applied to the generating function $E_k(\prod_i E(u_i)) = \prod_i E(u_i) T_k(\prod_i \prod_{a \in A} \frac{1+su_i x_a}{1+u_i x_a})$. $\square$

Remark 5.6. [Corollary 5.5](#) is an independent proof of Rick’s earlier all- k $e_k \star e_r$ rule; Hikita’s [8, Thm. 3.12] is the case $k = 1$. [\[TODO: Decide whether the Pieri package \(207b, two-column, \$\ell\$ -column rules\) gets a paragraph; v7.5 scope omits it.\]](#)

6 Second order: every valuation-2 lead

By [Theorem 4.5](#), valuation-2 leads reduce to connected leads with $\ell(\lambda) = 3$, $\kappa = 1$; by [Corollary 5.2](#) to two-part μ . Put $T_ag := \sum_{|A|=a} c_A X_A g(X_A)$ for $g \in \Lambda_a$, and

$$\Gamma_a(e_b, e_c) := E_a^{(2)}(e_b e_c) - e_b E_a^{(2)}(e_c) - e_c E_a^{(2)}(e_b) = \sum_{1 \leq r \leq b, 1 \leq q \leq c} (-1)^{r+q} e_{b-r} e_{c-q} T_a(p_r p_q).$$

Proposition 6.1. *If (a, b, c) is any ordering of λ and $\kappa(\lambda, \mu) = 1$, then $[(s-1)^2]c_{\lambda\mu} = [e_\mu](\Gamma_a(e_b, e_c) + D_a D_b(e_c))$.*

Theorem 6.2 (Constant-term adjoint formula). *For $g \in \Lambda_a$, $F \in \Lambda$: $\varphi_a(t) \langle T_ag, F \rangle_t = \text{CT}[Z g(z) F(z_1^{-1}, \dots, z_a^{-1}) K(z)]$, where $Z = z_1 \cdots z_a$ and $K(z) = \prod_{1 \leq i < j \leq a} \frac{z_j - z_i}{z_j - tz_i}$, expanded in z_i/z_j ($i < j$).*

Proof idea. T_ag restricted to a variables is Zg and lies in $\text{span}\{P_\mu : \ell(\mu) = a\}$. Evaluate $\langle G, q_\alpha \rangle_t = [y^\alpha]G$ via Cauchy, write $\langle G, Q_\lambda \rangle_t$ as a constant term by the raising-operator formula, and symmetrise $K = \Delta\kappa$ with $\sum_w w(\kappa) = v_a(t)$. (Ingredients: [12, Ch. III, §§1–2 and §4].) **[TODO: Equation-level locators deliberately omitted until checked against the book.]** This is Jing's vertex-operator calculus [9] with the vertex operators hidden. $\square$

Theorem 6.3 (Two-point formula). *Let $g \in \Lambda_a$ be homogeneous of degree d , $n = a + d$, $x, y \geq 1$, $x + y = n$, and $\Phi_a(g; x, y) := \langle T_ag, p_x p_y \rangle$. For $1 \leq A \leq a - 1$ put $B = a - A$ and $G_A(w) = \sum_k G_{A,k} w^k := g(1, t, \dots, t^{A-1}, w, wt, \dots, wt^{B-1})$. Then*

$$\Phi_a(g; x, y) = (-1)^a (1 - t^x)(1 - t^y) \left\{ \sum_{A=1}^{a-1} t^{-AB} \left[\frac{G_{A,y-B}}{(1-t^A)(1-t^B)} + \frac{1}{1-t^a} \sum_{m \geq 1} (t^{-Am} - t^{Bm}) G_{A,y-B-m} \right] - \frac{g(1, t, \dots, t^{a-1})}{1-t^a} \sum_{j=0}^{a-1} t^{-jy} \right\}.$$

Proof idea. Insert $S(u) = \sum_i 1/(uz_i - 1)$ into [Theorem 6.2](#) and take iterated residues: every variable lands on a t -string $1/u, t/u, \dots$ or $1/v, t/v, \dots$. A string of length A with base q has weight $(-1)^{A-1} \varphi_{A-1} q^A$, and two strings interact through the shuffle identity

$$\text{Sh}_{A,B}(w) = \begin{bmatrix} A+B \\ A \end{bmatrix}_t t^{-AB} \frac{(1-w)(1-t^{B-A}w)}{(1-t^{-A}w)(1-t^Bw)}, \quad w = u/v,$$

proved by a last-letter recursion. Take Taylor coefficients at $w = 0$. $\square$

Remark 6.4 (Relation to Green polynomials). Write $p_\mu = \sum_\lambda X_\mu^\lambda(t) P_\lambda(x; t)$. For $g = P_\rho$ and $\lambda = \rho + 1^a$ one has $\Phi_a(P_\rho; x, y) = (1 - t^x)(1 - t^y) X_{(x,y)}^\lambda(t) / b_\lambda(t)$, so [Theorem 6.3](#) evaluates Green polynomials at two-part classes; the two t -strings are the Frobenius orbits of

a maximal torus of type (x, y) [6]. [Theorem 6.2](#) is Jing's constant-term expression for the matrix element $\langle Q_\lambda, p_\mu \rangle$ [9], cf. [10, (2.17), (2.20)]. Jing and Liu [10, Thm. 2.7] expand the same matrix element as a nested sum valid for all μ , extracting one variable at a time; at two-part μ that sum does not close up, since its inner sums run over classes of arbitrary length. Their closed forms [10, Thm. 2.10, (2.37)–(2.40)] restrict the Hall–Littlewood index λ , not the class μ ; the case $\ell(\lambda) = 2$ goes back to Morris [14].¹ We evaluate by residues instead, and at $\ell(\mu) = 2$ the poles organise into two t -strings. The formula is explicit, with no sum over classes, and linear in the two-string specialisation G_A of g ; for general ρ , G_A is itself a finite (branching) sum, so [Theorem 6.3](#) is not a product formula.

Example 6.5 (Two rows). For $a = 2$, only $A = B = 1$ occurs, and $G_1(w) = P_\rho(1, w; t) = w^{\rho_2} \frac{(1-tw) - w^m(w-t)}{1-w}$ with $m = \rho_1 - \rho_2 \geq 1$ ($G_1 = w^{\rho_2}$ if $m = 0$). Substituting into [Theorem 6.3](#) gives, for $\lambda = (\lambda_1, \lambda_2)$ and $y \leq x$,

$$X_{(x,y)}^{(\lambda_1, \lambda_2)}(t) = (t-1) t^{\lambda_2-1-y} (1+t^y) \quad (y < \lambda_2), \quad m_{xy} - (1-t) t^{\lambda_2-1} \quad (y = \lambda_2), \quad (t-1) t^{\lambda_2-1} \quad (y > \lambda_2)$$

in agreement with an independent Kostka–Foulkes computation for $|\lambda| \leq 10$. This is the two-part-class case of the range $\ell(\lambda) = 2$ treated by Morris.

Theorem 6.6 (Closed $\ell = 3$, $\kappa = 1$ leads). *Let $\lambda = (a, b, c)$ be any ordering of a length-3 partition, $\mu = (x, y)$ with $\kappa(\lambda, \mu) = 1$, $n = x + y$, $m_{xy} = 2$ if $x = y$ and 1 otherwise. Put $\Lambda_a(r, q) := \text{lin}_e T_a(p_r p_q) = (-1)^{r+q} \frac{[a+r+q]}{[a]} [a]_{t^r} [a]_{t^q}$ and $U_a(b, c; x, y) := ((-1)^n \Phi_a(p_b p_c; x, y) - \Lambda_a(b, c)) / m_{xy}$. Then*

$$\begin{aligned} [(s-1)^2] c_{\lambda\mu} = & (-1)^{b+c} U_a(b, c; x, y) + \sum_{\substack{1 \leq r < b \\ \{b-r, a+r+c\} = \{x, y\}}} (-1)^{r+c} \Lambda_a(r, c) \\ & + \sum_{\substack{1 \leq q < c \\ \{c-q, a+b+q\} = \{x, y\}}} (-1)^{b+q} \Lambda_a(b, q) + [e_x e_y] D_a(M_{bc}), \end{aligned}$$

with D_a the derivation $D_a(e_j) = M_{aj}$ of [Proposition 3.3](#).

Proof idea. [Proposition 6.1](#); in the expansion of Γ_a only terms with $r = b$ or $q = c$ survive $[e_x e_y]$; single- e parts come from [Lemma 3.9](#), and $\langle G, p_x p_y \rangle = (-1)^n (m_{xy} [e_x e_y] G + \text{lin}_e G)$ converts [Theorem 6.3](#) into U_a . $\square$

Corollary 6.7. *Every lead at valuation 2 ($\kappa = \ell - 2$, any ℓ) has a closed formula: by [Theorem 4.5](#) it is a sum of products of pair leads ([Proposition 3.3](#)) and $\ell = 3$, $\kappa = 1$ leads ([Theorem 4.1](#), [Corollary 5.2](#), [Theorem 6.6](#)).*

Example 6.8. $(3, 3, 3) \rightarrow (7, 2)$: $\text{Lead} = 2t^{13} + 3t^{12} + 3t^{11} + 6t^{10} + 6t^9 + 6t^8 + 9t^7 + 6t^6 + 3t^5 + 9t^4 + 6t^3 + 3t + 4$ (agrees with direct computation). Leads are *not* positive in general: $(4, 4, 2) \rightarrow (7, 3)$ has $\text{Lead} = (t+1)(t^2+1)(2t^8+t^7+t^6+3t^4+t^3+t^2-t+2)$.

¹We have not seen [14] first-hand; its scope is as described in [10, p. 12, before Thm. 3.2].

Computer checks. All statements in this section were checked against an independent implementation of Hikita's subset formula: [Theorem 6.6](#) on all 16 class-4 pairs with $n \leq 12$ and on 27 pairs in all 3 orderings with $n \leq 10$; [Theorem 6.3](#) in 229 cases ($a \leq 5$); and the Green-polynomial dictionary symbolically for $|\lambda| \leq 8$ and numerically for $|\lambda| \leq 9$.

7 Open problems

1. **Valuation ≥ 3 .** Connected leads with $\ell \geq 4$, $\kappa = 1$. Hierarchy: $v = 1 \leftrightarrow p_n \leftrightarrow$ one t -string ([Lemma 3.9](#)); $v = 2 \leftrightarrow p_x p_y \leftrightarrow$ two strings ([Theorem 6.3](#)). Does the three-string shuffle sum factor pairwise?
2. **Zero set at generic t .** At $t = -2$ seven pairs with $n = 6$ have $v > \ell - \kappa$; for coarsenings this is cancellation between histories, e.g. $\text{Lead}_{(1,1,1),(3)} = [3]_t(2+t)$. Describe the zero set of $[(s-1)^{\ell-\kappa}]c_{\lambda\mu}$ in t .
3. **Counting.** A formula for $N(\lambda, \mu)$ in [Theorem 3.4](#); the observed $\text{Lead}_{\lambda, (n-1,1)}(1) = 2n^2 - 6n + 3$ (computed only).
4. **Positivity.** Leads are not t -positive (Example above), so no “ t -count of minimal flows” can hold. Is there a signed or cancellation-free model?
5. **Cumulant link.** Dołęga's “missing link” between Macdonald cumulants and shuffle-type polynomials [4]: is the $\star$ -lead (a cumulant of $e_k \mapsto t^{\binom{k}{2}}$) part of it?

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[TODO: Robin; Clio (peer agent) for reviews; per Robin.]

References

- [1] J. Désarménien, B. Leclerc, and J.-Y. Thibon. “Hall–Littlewood functions and Kostka–Foulkes polynomials in representation theory”. *Sém. Lothar. Combin.* **32** (1994). Art. B32c; TODO verify.
- [2] P. Di Francesco and R. Kedem. “Difference equations for graded characters from quantum cluster algebra”. *Transform. Groups* **23.2** (2018). Locators refer to arXiv v2: eq. (5.15), Cor. 5.8 (display (5.18)), eq. (5.25), Cor. 5.18 (eq. (5.27)), pp. 391–424. [arXiv:1505.01657](#).
- [3] P. Di Francesco and R. Kedem. “ (t, q) -deformed Q-systems, DAHA and quantum toroidal algebras via generalized Macdonald operators”. *Comm. Math. Phys.* **369** (2019). TODO verify title, pp. 867–928. [arXiv:1704.00154](#).

- [4] M. Dołęga. “Macdonald cumulants, G -inversion polynomials and G -parking functions”. *European J. Combin.* **75** (2019). TODO verify title, pp. 172–194. [arXiv:1707.02656](#).
- [5] I. M. Gessel and B. E. Sagan. “The Tutte polynomial of a graph, depth-first search, and simplicial complex partitions”. *Electron. J. Combin.* **3.2** (1996). Research Paper 9.
- [6] J. A. Green. “The characters of the finite general linear groups”. *Trans. Amer. Math. Soc.* **80** (1955), pp. 402–447.
- [7] J. Haglund and V. Tewari. “Theta operators at $t = 1$, Macdonald cumulants, and LLT positivity”. TODO verify title. 2026. [arXiv:2609.29957](#).
- [8] T. Hikita. “ (q, t) -Chromatic Symmetric Functions”. Def. 3.4, Thm. 3.12, Prop. 3.8, Cor. 3.10 used. 2025. [arXiv:2503.23597](#).
- [9] N. Jing. “Vertex operators and Hall–Littlewood symmetric functions”. *Adv. Math.* **87** (1991), pp. 226–248.
- [10] N. Jing and N. Liu. “The Green polynomials via vertex operators”. 2021. [arXiv:2104.04411](#).
- [11] A. N. Kirillov. “New combinatorial formula for modified Hall–Littlewood polynomials”. §3.2 used: the $e \rightarrow P$ transition matrix $M(e, P)$; $R_{\lambda\mu}(1)$ counts 0-1 matrices (Knuth). 1998. [arXiv:math/9803006](#).
- [12] I. G. Macdonald. *Symmetric Functions and Hall Polynomials*. 2nd ed. Oxford University Press, 1995.
- [13] C. L. Mallows and J. Riordan. “The inversion enumerator for labeled trees”. *Bull. Amer. Math. Soc.* **74** (1968), pp. 92–94.
- [14] A. O. Morris. “A survey on Hall–Littlewood functions and their applications to representation theory”. *Combinatoire et représentation du groupe symétrique (Strasbourg, 1976)*. Vol. 579. Lecture Notes in Math. Springer, 1977, pp. 136–154. [DOI](#).
- [15] O. Penrose. “Convergence of fugacity expansions for classical systems”. *Statistical Mechanics: Foundations and Applications*. TODO verify; read only via Procacci–Yuhjtman [arXiv:1508.07379](#). 1967.

AI disclosure

[**TODO:** Robin to edit wording; mandatory, uncapped, not counted toward 12 pages. Also fill in the SoftConf form field.]

The mathematics in this extended abstract was produced by an AI agent. “Rick” is an autonomous AI research agent built on Anthropic’s Claude models [**TODO: exact model names and versions**]. Rick chose the questions, ran the computer experiments, conjectured the statements, found and wrote the proofs, carried out the literature/novelty searches, and drafted this text. This includes exploration, coding, computation, mathematical reasoning, proving and writing.

- **Statements and proofs:** all theorems in Sections 2–5 (including the block valuation law, the merge-weight formula, the full-merge and block-multiplicative lead formulas, the Box Complement theorem, the constant-term and two-point formulas, and the valuation-2 leads) were conjectured and proved by the AI agent. [**TODO: State which proofs have been checked by a human, if any; currently: none fully.**]
- **Verification:** computer checks (SymPy/Python, written by the AI agent) against an independent implementation of the subset formula; a second AI agent (“Clio”) reviewed some, not all, of the inputs. [**TODO: list which.**]
- **Literature:** prior-art searches were performed by the AI agent and its sub-agents; identified prior art is cited in the text.
- **Human role:** [**TODO: Robin: describe your role (e.g. supervision, submission, verification).**]